Y PAVAN KUMAR

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 2 years of professional experience delivering production-grade systems for DRDO government clients. Specialises in building scalable web applications and real-time industrial monitoring solutions using Python, Django, React, Next.js, and TypeScript. Track record of eliminating manual processes, achieving 99% accuracy in barcode tracking, and automating 85% of CNC machine data entry. Adept at end-to-end delivery — from database design and REST API development to containerised deployment with Docker.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** React, Next.js, Django, Node.js, Bootstrap, Tailwind CSS
- **Databases:** PostgreSQL, MySQL
- **Tools & DevOps:** Git, Docker, REST API, PyInstaller, NextAuth.js, Serial Communication
- **Testing & Methodology:** API testing, Debugging, Hardware-Software Integration
- **Other:** EAN-13 Barcode, DataTables, Gantt Charts, OEE Metrics, Role-Based Access Control

WORK EXPERIENCE

Software Engineer | **Techfluent Solutions** *Mar 2024 – Present · Hyderabad, India*

- **TraceBar Inventory System (Client: LRDE DRDO):** Architected and shipped a full-stack supply chain platform managing supplier onboarding, receipt processing, item issues, and repair tracking — eliminating 90% of manual tracking errors.
- Implemented EAN-13 barcode generation and scanning for product receipts, achieving **99% scan accuracy** and removing all manual data entry from the receiving workflow.
- **CNC OEE Monitor (Client: MTPF ODF):** Designed a real-time Overall Equipment Effectiveness dashboard consuming REST API data from CNC machines, reducing manual reporting effort by **85%** and enabling live downtime analytics.
- **EasyDAQ (Client: KITS Warangal):** Built a Python desktop data acquisition application with serial port communication supporting LED control, voltage/resistance measurement, power source control, logic analyser, and 3-channel oscilloscope. Packaged as standalone .exe via PyInstaller.
- **CRM (Client: DRDL DRDO):** Delivered a full-stack Customer Relationship Management system (Next.js, React, TypeScript, MySQL) with session auth, role-based access, enquiry pipeline, proposal/order management, PDF/Excel export, and real-time notifications.
- **QMS (Client: CABS DRDO):** Built a Quality Management System (Next.js, TypeScript, PostgreSQL) with 8 role-based workflows, inspection request lifecycle, multi-stage approvals, inspection checks, and a complete audit trail.
- **PMS:** Developed an enterprise Project Management System featuring SVAR Gantt charts, task dependencies, resource management, and executive reporting.

EDUCATION

Bachelor of Engineering — Information Technology *2020 – 2023*
MVSR Engineering College, Hyderabad

Diploma — Computer Science Engineering *2016 – 2020*
Government Institute of Electronics, Secunderabad

CAREER BREAK

Self-directed Learning & Skill Development *2023 – 2024*

Dedicated period post-graduation to mastering full-stack technologies — React, Next.js, Django, PostgreSQL, REST APIs, and Docker — before joining professional employment.

KEY ACHIEVEMENTS

- Reduced inventory tracking errors by **90%** through EAN-13 barcode integration for a DRDO defence lab
- Automated CNC machine OEE monitoring, eliminating **85%** of manual data entry at a government ordnance factory
- Delivered **7+ production-ready applications** across web and desktop platforms for DRDO, KITS, and enterprise clients
- Designed and shipped a multi-role QMS with 8 workflow stages used by CABS DRDO quality inspection teams

ADDITIONAL INFORMATION

- **Relocation:** Open to relocating anywhere within India
- **Work Mode:** On-site, Hybrid, or Remote
- **Notice Period:** Immediate joiner / 30 days
- **Languages:** English (Fluent) · Hindi (Fluent) · Telugu (Native)